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**Point-of-Sale (POS) Technology in Tourism-Related Service
Environments:
A Comparative Analysis of McDonald's and PETRONAS**

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1.0 Introduction

The global tourism industry has undergone a profound digital transformation, with Point-of-Sale (POS) systems emerging as one of its most critical enabling technologies. Once functioning merely as cash registers, POS systems now serve as centralised hubs within the digital tourism ecosystem managing payment processing, customer data, loyalty programme integration, and operational analytics across hospitality and mobility environments (Springer Nature, 2024). Whether a tourist is dining at a quick-service restaurant, refuelling on a highway, or checking into a hotel, POS systems govern the speed, accuracy, and quality of these service interactions.

The shift toward cashless travel has accelerated this role considerably. A 2024 PaynoPain survey found that 85% of travellers prefer contactless payment methods, with 50% rating cashless access as very important during travel. According to the Mastercard Economics Institute's Travel Trends 2024 report, digital payments powered record-breaking global leisure travel spending reflecting how deeply embedded cashless POS infrastructure has become in the modern travel journey (SimPaisha, 2024). Nearly 90% of Gen Z and Millennial travellers now use mobile wallets abroad, creating strong consumer pressure on tourism-related service providers to deploy reliable, feature-rich POS platforms (PaynoPain, 2024).

This report examines POS technology adoption across two distinct yet tourism-relevant organisations: McDonald's, the world's largest quick-service restaurant (QSR) chain, and PETRONAS, Malaysia's national petroleum company with the country's largest fuel retail network. McDonald's functions as a hospitality and dining touchpoint for travelling consumers, while PETRONAS serves as a critical mobility and travel facilitation service provider. This report aims to compare and analyse how both organisations utilise POS technology to enhance customer experience and operational efficiency within tourism-related service environments, and to identify which demonstrates more effective technology integration.

2.0 Organisation A: McDonald's

2.1 Background

McDonald's Corporation is the world's largest quick-service restaurant chain, operating over 40,000 restaurants across more than 100 countries and serving approximately 69 million customers daily (McDonald's Corporation, 2023). Founded in 1940 in San Bernardino, California, McDonald's ranks among the world's top five most valuable brands globally. Within the tourism context, its global standardisation of menus, pricing, and service processes makes it a familiar and reliable hospitality touchpoint for international travellers navigating unfamiliar destinations. The brand's 'Accelerating the Arches' digital growth strategy places POS technology, loyalty infrastructure, and digital ordering at the centre of its competitive model.

2.2 POS Technology Use

McDonald's POS ecosystem integrates self-ordering kiosks, a mobile application, loyalty interfaces, drive-thru automation, and cloud-based analytics into a unified digital infrastructure. Over 130,000 self-ordering kiosks globally allow customers to browse, customise, and pay independently (Purohit et al., 2025). In 2024, McDonald's deployed a universal operating system across all customer-facing platforms built in partnership with Google Cloud enabling consistent, rapid rollout of innovations across thousands of outlets (QSR Magazine, 2023). AI-driven data from millions of daily transactions powers personalised promotions, menu recommendations, and predictive supply chain tools. The MyMcDonald's Rewards loyalty programme counts over 185 million active members across 60 markets, with U.S. members visiting 26 times per year post-enrolment compared to 10.5 times before demonstrating the measurable behavioural impact of POS-loyalty integration (Restaurant Technology News, 2025). McDonald's targets 250 million loyalty users and \$45 billion in annual digital sales by 2027 (McDonald's Corporation, 2023). Compared to PETRONAS, McDonald's POS places significantly stronger emphasis on personalisation, upselling, and CRM.

2.3 Benefits

McDonald's POS integration has delivered measurable commercial and experiential gains. Self-ordering kiosks increase average order values by 11-30% through AI-driven upsell prompts and extended digital browsing (Purohit et al., 2025). Order accuracy and perceived wait times have improved substantially, while the loyalty-POS

feedback loop delivering personalised offers tied to transaction records strengthens repeat engagement among business and leisure travellers. For tourism operators globally, McDonald's demonstrates how a well-integrated POS ecosystem can simultaneously serve as a revenue tool, a customer engagement platform, and a brand consistency mechanism.

2.4 Challenges

Despite its sophistication, McDonald's POS ecosystem faces serious reliability and security concerns. A 2025 Canopy survey found that 60% of kiosk users encountered problems, with 80% citing frozen screens, malfunctioning printers, or unreliable card readers, and over 75% reporting difficulties completing contactless payments (TheStreet, 2025). Cybersecurity incidents including a 2025 breach that exposed data from up to 64 million applicants highlight the tension between rapid digital innovation and security governance (Breached.company, 2025). The digital divide also poses a challenge: elderly tourists and international visitors unfamiliar with local language interfaces may struggle with kiosk navigation, undermining the inclusive hospitality experience McDonald's aims to deliver.

3.0 Organisation B: PETRONAS

3.1 Background

Petroleum Nasional Berhad (PETRONAS), established in 1974 and wholly owned by the Malaysian government, operates the country's largest fuel retail network through its subsidiary PETRONAS Dagangan Berhad (PDB), with over 1,000 stations nationwide (PETRONAS Dagangan, 2023). Ranked 216th on the Fortune Global 500, PETRONAS has a presence in over 100 countries. Within the digital tourism ecosystem, its petrol stations function as critical rest stops and refuelling touchpoints along Malaysia's highway network essential service environments where domestic and international road travellers regularly interact with digital payment infrastructure. The rise of cashless travel expectations has made PETRONAS's POS strategy directly relevant to overall traveller journey quality.

3.2 POS Technology Use

PETRONAS's POS strategy is anchored by Setel, Malaysia's first mobile application for pump-side fuel payment, launched in July 2018. Setel enables customers to select a pump, pay digitally, and complete refuelling without leaving their vehicle or handling cash (F&L Asia, 2020). The platform bundles payment, Mesra Rewards loyalty, parking, EV charging, motor insurance, and road tax renewal into a single digital mobility companion. In October 2023, PETRONAS relaunched a cardless Mesra Rewards programme allowing customers to earn points by entering only their phone number eliminating physical cards and offering Setel users up to triple the standard points rate, with redemptions including hotel stays and travel insurance (PETRONAS Dagangan, 2023). In June 2025, Setel integrated into the Touch 'n Go (TNG) eWallet as a mini programme, extending cashless fuel payments to millions of existing TNG users across 1,000+ stations without a separate app download (Setel, 2025). Setel Lite a streamlined, low-data version was introduced to serve rural users and older devices (The Edge Malaysia, 2025). Unlike McDonald's, PETRONAS prioritises transaction speed and mobility convenience, reflecting the brief, transactional nature of the fuel retail service interaction.

3.3 Benefits

Setel has delivered significant efficiency and sustainability gains. By enabling payment directly at the pump, the platform has reduced queue lengths, improved forecourt traffic flow, and decreased vehicle idling lowering both wait times and

emissions (The Edge Malaysia, 2025). Digital e-receipts replace paper, supporting PETRONAS Dagangan's environmental objectives. Geolocation integration guides travellers to the nearest PETRONAS station, functioning as an active smart mobility tool within the tourism ecosystem. The cardless Mesra relaunch strengthened loyalty among five million members by simplifying enrolment and broadening redemption options to include tourism-relevant products such as hotel stays and travel insurance (PETRONAS Dagangan, 2023). The TNG integration further lowered the digital adoption barrier, reinforcing PETRONAS's position as a cashless mobility leader in Malaysia (Setel, 2025).

3.4 Challenges

PETRONAS faces ongoing adoption and reliability challenges with the Setel platform. User reviews indicate recurring transaction auto-capture failures and a lack of clear in-app financial statements, which can erode trust among first-time cashless users (Apple App Store, 2024). Cybersecurity risk is explicitly identified in PETRONAS Dagangan's 2023 Integrated Report as a top-priority concern, with ongoing investments in cloud security and staff cyber safety training needed to protect the growing digital infrastructure (PETRONAS Dagangan, 2023). The digital divide remains acute in rural areas and among older travellers: while Setel Lite partially addresses connectivity limitations, customers without smartphones continue to depend on traditional cashier-based POS terminals, constraining the universality of PETRONAS's cashless service vision.

4.0 Comparison Table

Technology: Point-of-Sale (POS) System

Comparison Criteria	Organisation A: McDonald's	Organisation B: PETRONAS (Setel)
A. Main Features / Functions	- 130,000+ self-ordering kiosks globally - Mobile app with AI personalisation - Unified Google Cloud OS (2024) - CRM-integrated MyMcDonald's Rewards - Drive-thru automation & digital ordering	- Mobile pump-and-pay via Setel app - TNG eWallet integration (1,000+ stations) - Cardless Mesra Rewards loyalty (2023) - EV charging, parking, insurance in-app - Setel Lite for low-connectivity users
B. Customer / User Experience	- AI personalised menu recommendations - Reduced queues via kiosk ordering - Multiple payment methods supported - Gamified loyalty with frequent rewards - 185M+ active loyalty members globally	- Contactless fuelling without leaving vehicle - Earn up to 3x Mesra points via Setel - Digital e-receipts (paperless) - Geolocation station finder for travellers - Rewards include hotel stays & travel insurance
C. Adoption Issues	- 80% of kiosk users report technical issues - 75%+ payment reliability concerns - Digital divide for elderly/non-tech tourists - Cybersecurity vulnerabilities in kiosks - Learning curve for first-time users	- Transaction auto-capture failures reported - Requires smartphone & internet access - Digital divide in rural/remote areas - App onboarding complexity for new users - Dependency on mobile data connectivity
D. Marketing Impact	- Target: 250M loyalty users by 2027 - \$45B annual systemwide digital sales target - AI-driven personalised push promotions - Globally ranked top-5 most valuable brand - 'Feel-Good Marketing' digital-first strategy	- Mesra Bonanza 2023 (Visa collaboration) - Grand prize: all-expenses-paid Paris trip - 5M+ Mesra loyalty programme members - TNG partnership extends reach to millions - Cashless mobility leader in Malaysia
E. Sustainability / Future Potential	- Google Cloud AI for supply chain efficiency - GenAI for operational optimisation - Expanding to 50,000 restaurants by 2027 - Reduced food waste via predictive analytics - Trajectory toward fully	- Digital receipts reduce paper waste - EV charging supports green travel mobility - Setel Lite promotes digital inclusion - Expanding into insurance & road tax renewal - Aligned with Malaysia's

Comparison Criteria	Organisation A: McDonald's	Organisation B: PETRONAS (Setel)
	cashierless outlets	cashless economy vision

5.0 Discussion

The comparative analysis reveals two organisations that have invested significantly in POS technology but have oriented their systems toward fundamentally different strategic goals. McDonald's has built a data-intensive, hospitality-focused engagement platform, while PETRONAS has developed a lean, speed-optimised mobility payment ecosystem. Understanding why these differences exist and what they mean for tourism service quality is the central analytical task of this discussion.

Both organisations demonstrate strong alignment with the cashless travel expectations of modern tourists. McDonald's 185 million active loyalty members visiting 2.5 times more frequently after enrolment illustrate the commercial and behavioural power of a tightly integrated POS-loyalty platform (Restaurant Technology News, 2025). PETRONAS's Mesra Rewards integration of hotel stays and travel insurance as redemption options signals a deliberate strategy to embed the brand within the broader tourist journey rather than limiting its value to the fuelling transaction. Both also share vulnerabilities: the 80% kiosk failure rate at McDonald's (TheStreet, 2025) and Setel's recurring transaction reliability issues both highlight that POS performance problems have immediate consequences for tourist satisfaction particularly among travellers who may only experience the system once.

The most significant difference lies in the depth and scope of each system's role in the tourist experience. McDonald's POS creates experience value: AI personalisation, gamified loyalty, and upsell-driven engagement enrich each transaction and encourage repeat interaction over time. PETRONAS's Setel creates efficiency value: frictionless, vehicle-based payment that optimises the brevity of the fuel retail interaction. This distinction is analytically important because it reflects the different service environments each organisation operates within: diners expect engagement and customisation, while road travellers expect speed and reliability above all else. Crucially, PETRONAS's Setel Lite and TNG eWallet integration demonstrate a stronger commitment to digital inclusion, actively reducing barriers for older travellers and low-connectivity users a dimension where McDonald's kiosk-centric model is relatively weaker.

Cybersecurity represents the most critical shared challenge. Both organisations accumulate sensitive customer payment data at scale, and both have acknowledged

significant security risks McDonald's through actual breaches (Breached.company, 2025), and PETRONAS through explicit risk identification in its Integrated Report (PETRONAS Dagangan, 2023). Academic research supports this concern, noting that a single negative digital payment experience can permanently reduce consumer willingness to use a platform (Springer Nature, 2024). In evaluating overall effectiveness, McDonald's demonstrates superior performance in personalisation, loyalty depth, and revenue uplift, while PETRONAS leads in mobility-oriented cashless delivery and digital inclusion. For the specific context of tourism service quality where mobility is the foundational enabler of tourist movement PETRONAS's frictionless, ecosystem-integrated approach arguably delivers more immediate and universally accessible impact.

6.0 Conclusion and Recommendations

This report has demonstrated that POS technology constitutes a foundational element of the digital tourism ecosystem. McDonald's and PETRONAS exemplify two distinct and instructive approaches: McDonald's as a hospitality-driven personalisation platform, and PETRONAS as a mobility-centric smart service system. Both have achieved meaningful outcomes through their POS investments, yet both face persistent shared challenges in system reliability, cybersecurity governance, and digital inclusion that must be addressed to ensure POS technology continues to enable rather than frustrate the tourist experience.

The following targeted recommendations are proposed:

Recommendation 1 – McDonald's: Prioritise Kiosk Reliability

McDonald's should deploy real-time kiosk health monitoring using its existing Google Cloud infrastructure to enable predictive fault detection and faster maintenance response. A minimum 'uptime standard' should be embedded as a global franchise requirement, particularly for outlets in high-tourist-footfall locations.

Recommendation 2 – McDonald's: Enhance Digital Accessibility

Multilingual kiosk interfaces should be standardised globally, and dedicated staff support should be available during peak tourist seasons to assist elderly and international visitors ensuring digital innovation does not come at the cost of inclusive hospitality.

Recommendation 3 – PETRONAS: Improve Setel Transaction Transparency

PETRONAS should introduce a clear in-app transaction history feature with weekly and monthly statements. This directly addresses reported user concerns about financial transparency and will build trust among new digital payment adopters a critical audience for Malaysia's broader cashless economy agenda.

Recommendation 4 – PETRONAS: Expand Tourism Smart Hub Integration

PETRONAS should leverage its 1,000+ station network by integrating travel booking services, destination guides, and accommodation partnerships within Kedai Mesra outlets transforming petrol stations from mere refuelling stops into multi-service digital tourism touchpoints along Malaysia's key travel corridors.

Recommendation 5 – Both: Strengthen Cybersecurity Governance

Both organisations should commission independent cybersecurity audits of their POS

ecosystems, establish publicly communicated data breach response protocols, and invest in ongoing staff training. Transparent security governance is essential to maintaining consumer trust in cashless digital tourism infrastructure.

In conclusion, as digital payment technology continues to evolve, POS systems will remain central to the competitiveness of tourism-related service providers. The experiences of McDonald's and PETRONAS offer valuable, practical lessons for any organisation seeking to optimise the intersection of technology, customer experience, and tourism service excellence.

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